



FINAL PROJECT REPORT

2110215 Programming Methodology

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Tower Defense Project (JavaFX)

1. Game Overview & Manual (How to Play)

Objective

Defend your castle for 3 minutes while enemies continuously spawn and move along the generated path. You win if the timer reaches 0 and no enemies remain alive; you lose if castle HP reaches 0.

Initial resources:

- Base Health: 100
- Money: 500
- Timer: 03:00

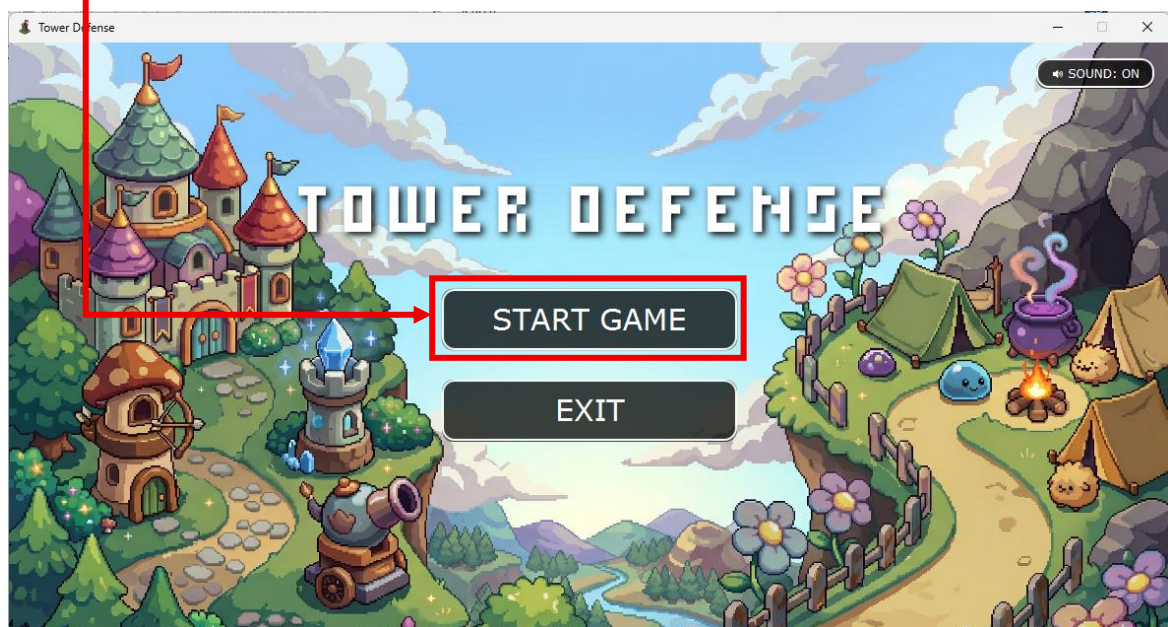


Win/Loss Conditions

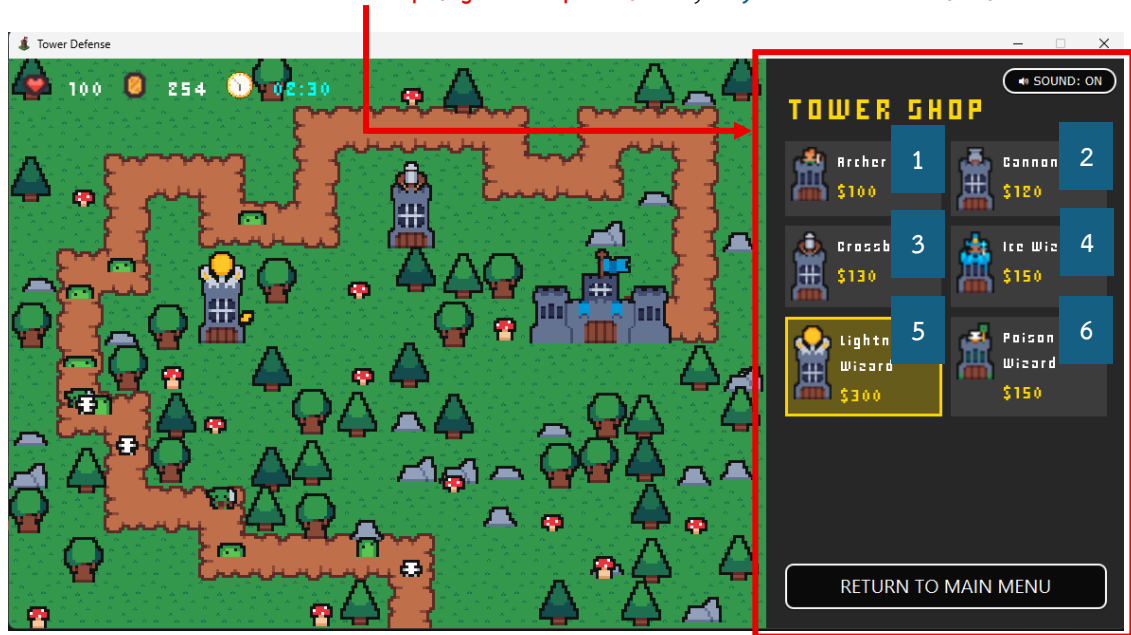
- **Loss:** in GameManager.handleEnemyReachedBase, enemy reaching end reduces base HP; when $HP \leq 0$, `isGameOver = true`.
- **Win:** in GameManager.checkAndHandleVictory, when timer expired, no active enemies remain, and base still alive, `isVictory = true`.

Core Gameplay Flow

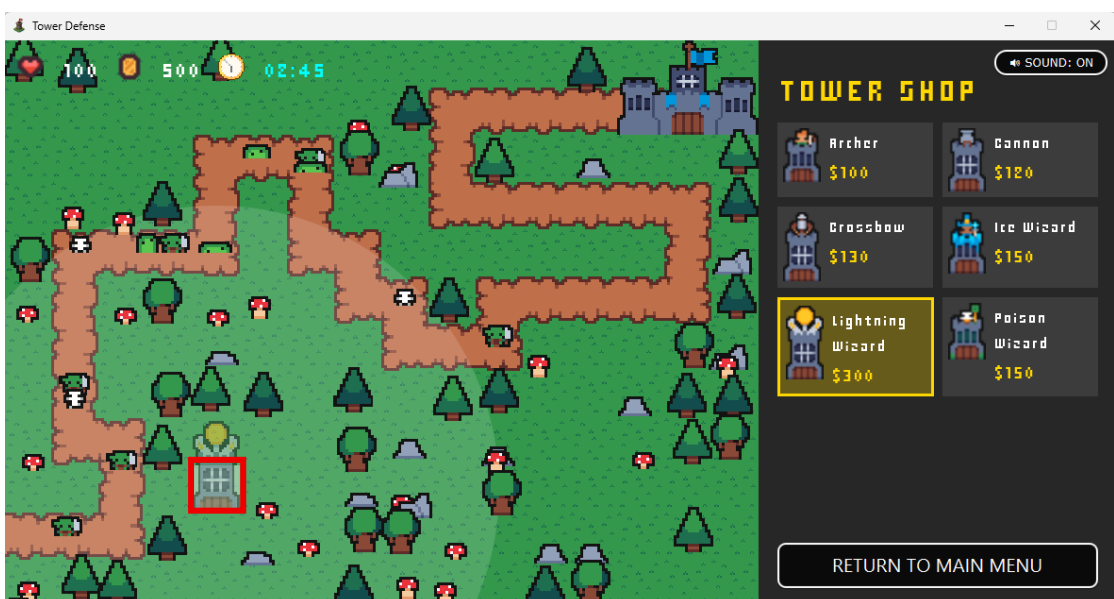
1. **Start** from the main menu (MainMenu) and enter gameplay (`Main.startGameFromMenu()`).



2. Select a tower from the **Tower Shop** (right-side panel) or by **keyboard shortcut (1-6)**.



3. **Place towers** on valid grass tiles (not path, not castle zone, not occupied by other towers, not blocked by decoration).



- Towers **auto-attack** enemies in range.



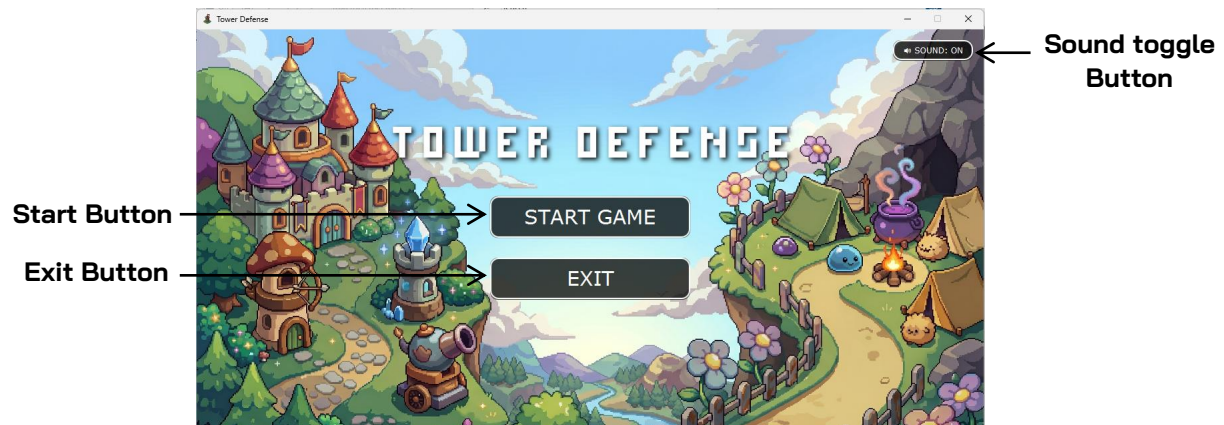
- Earn money from enemy kills.
- Click a placed tower to open **Tower Status**, then **upgrade** if affordable.



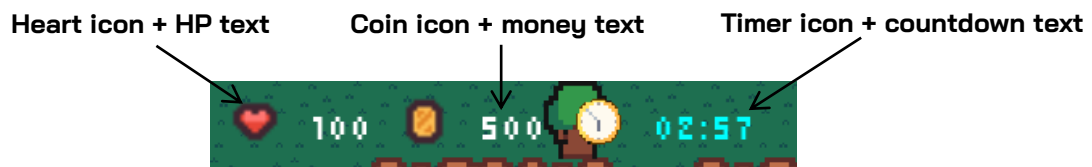
- Survive until time ends.

UI Components

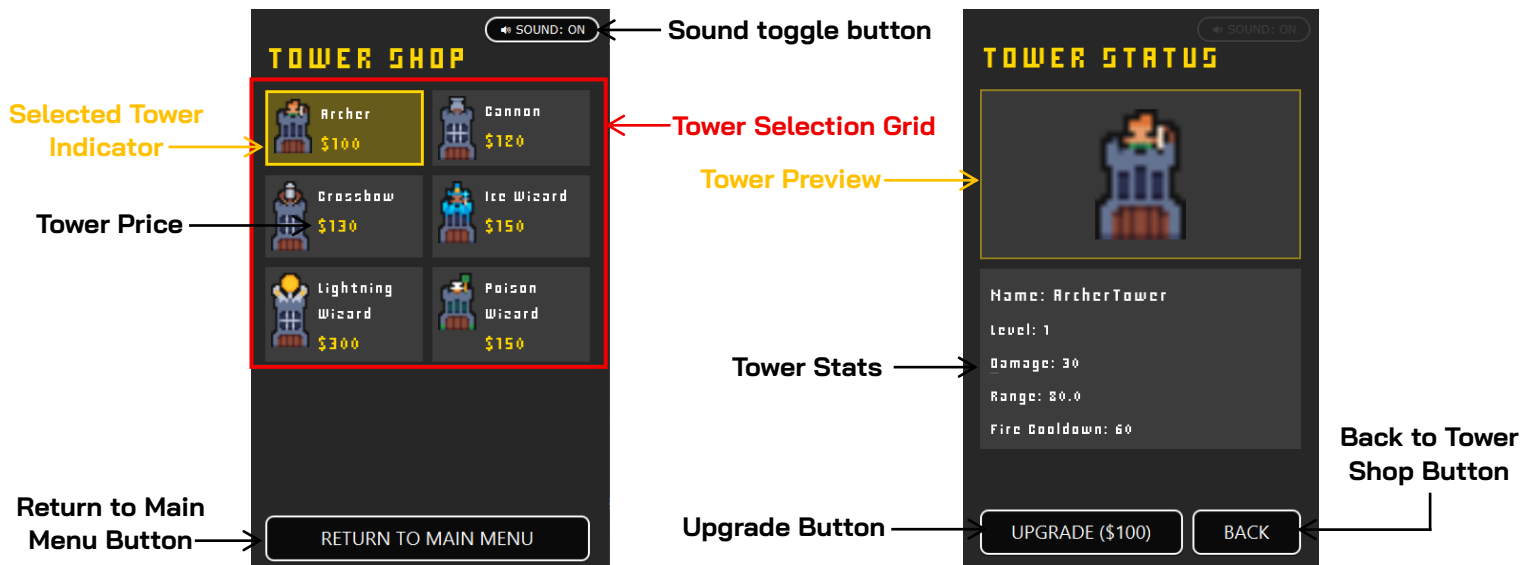
- Main Menu (MainMenu)



- HUD (top-left in Main)



- Side Panel

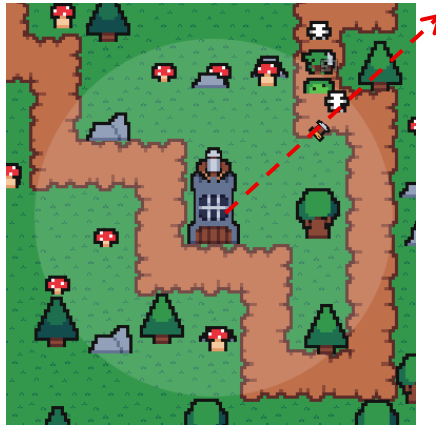


- In-game Visual Feedback

- Hover validity tile highlight & Tower range circles & Ghost tower preview



- Projectile rotation toward targets



- Enemy hit flash & Floating damage numbers (DamageText)

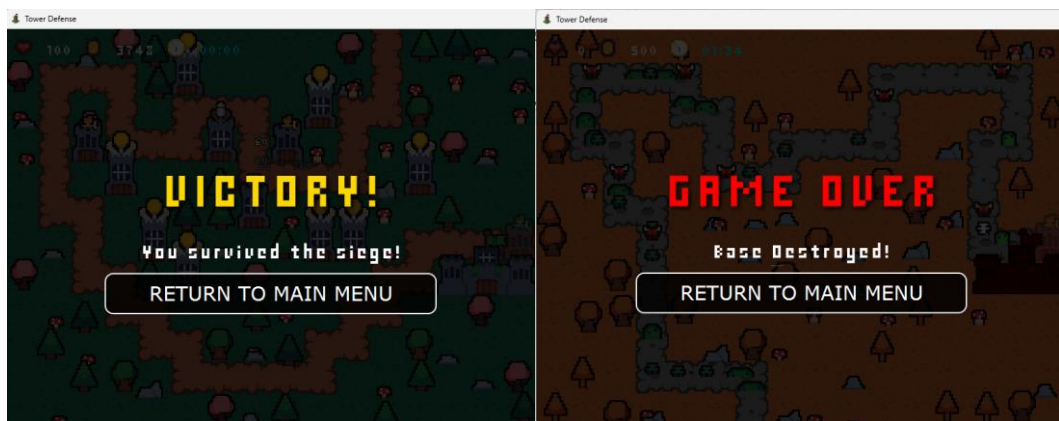


- Castle hit shake effect



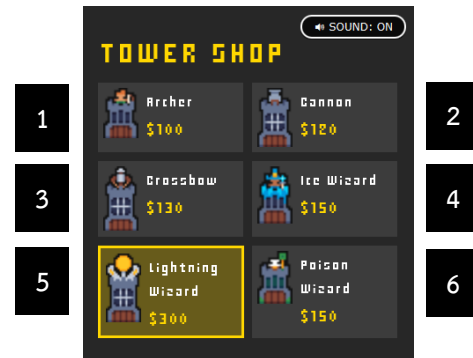
- Terminal Overlays

- Victory overlay & Game Over overlay



Controls

- **Left Click (Map)**
 - If clicking an existing tower: select/deselect tower and show range.
 - Otherwise: place currently selected tower.
- **Right Click (Map/Shop)**
 - clear selection/cancel placement mode.
- **Keyboard shortcuts (GameView.handleKeyPress)**
 - 1: Archer
 - 2: Cannon
 - 3: Crossbow
 - 4: Ice Wizard
 - 5: Lightning Wizard
 - 6: Poison Wizard



Placement Rules (from GameManager.canPlaceTower + GameMap.isBuildable)

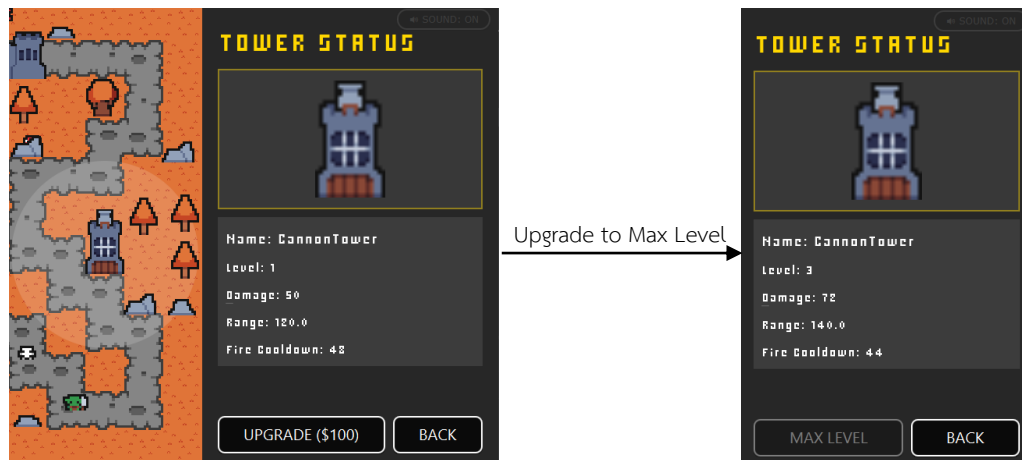
A tower can be placed only when all conditions are true

- A tower type is selected.
- Game is not over.
- Tile is inside grid.
- Tile value is buildable grass (0), not path (1) or castle (2).
- Tile is not inside castle clearance zone.
- Tile has no decoration (🌲🌳🌴🌵🏠🏡🏢🏣🏤🏥🏦🏧🏨🏩🏪🏫🏬🏭🏮🏰🏱🏲🏳🏴🏵🏶🏷🏸🏹🏺🏻🏼🏽🏾🏿🏠🏡🏢🏣🏤🏥🏦🏧🏨🏩🏪🏫🏬🏭🏮🏰🏱🏲🏳🏴🏵🏶🏷🏸🏹🏺🏻🏼🏽🏾🏿).
- Tile is not already occupied by another tower.
- Player has enough money for selected tower.



Tower Status & Upgrade System

Selecting a placed tower opens the **Tower Status Panel** (Main.createTowerStatusPanel) showing



Upgrade behavior (Tower.upgrade())

- level +1 (up to max level 3)
- Damage increases (scaled)
- Range +10
- Fire cooldown decreases (minimum-limited)
- Next upgrade cost increases

Tower Types (Actual Class Stats)

Tower Class	Cost	Damage	Range	Fire Cooldown
ArcherTower	100	30	80.0	60
CannonTower	120	50	120.0	48
CrossbowTower	130	35	150.0	40
IceWizardTower	150	60	150.0	120
PoisonWizardTower	150	60	150.0	120
LightningWizardTower	300	60	300.0	60

Enemy Types (Actual Class Stats)

Enemy Class	HP	Speed	Reward	Base Damage to Castle
SlimeEnemy	100	1.0	10	1
BatEnemy	50	2.5	15	1
BigSlimeEnemy	180	0.8	20	2
GoblinEnemy	90	1.8	18	2
SkeletonEnemy	130	1.2	16	3
ZombieEnemy	170	0.9	22	4
DemonEnemy	250	1.5	35	5
KingSlimeEnemy	320	2.0	45	50

2. Game Features & Complexity

Custom Graphics Pipeline

- Centralized texture preloading with AssetManager (environment, towers, projectiles, enemies).
- Theme-dependent map visuals (Theme.NORMAL, Theme.AUTUMN, Theme.SPRING).
- Depth-sorted rendering of decorations/enemies/towers for proper visual layering.

Audio Integration (SoundManager)

- Dedicated SFX for clicks, enemy hit, castle hit, victory, defeat.
- Separate menu and in-game looping BGM using MediaPlayer.
- Global mute toggle with UI synchronization.

Scaling Difficulty via Time-Based Waves

- GameManager changes enemy pools and spawn intervals by match phase
 - Early phase: slower spawn interval, easier enemies.
 - Mid phase: medium interval, stronger enemies.
 - Late phase: fastest interval, strongest enemies.

Fixed-Time Step Game Loop

- Main.createGameLoop() uses an accumulator with fixed logic step
 - LOGIC_TICKS_PER_SECOND = 60
 - LOGIC_STEP_SECONDS = 1.0 / 60
 - bounded MAX_LOGIC_STEPS_PER_FRAME
- important for stable simulation, deterministic update rhythm, reduced frame-rate dependency.

Procedural Path Generation + Pathfinding

- PathGenerator.generateRandomPath() creates randomized valid maps.
- GameMap.generatePath() reconstructs route to castle via BFS.
- Combined with tower-placement constraints, this increases replayability and algorithmic depth.

3. Overall Architecture & Design

application package (View/Presentation Layer)

- JavaFX bootstrapping and scene management (Main, Launcher, MainMenu)
- Rendering and player input (GameView)
- UI helpers and assets/audio (UIUtils, AssetManager, SoundManager)

logic package (Core Game Domain)

- Match state orchestration (GameManager)
- Combat entities (logic.tower.*, logic.enemy.*)
- Map/path systems (logic.map.*)
- Interfaces/contracts (logic.interfaces.*)

Why this separation matters?

- Easier maintenance and debugging (UI issues vs gameplay issues are isolated).
- Better testability (core mechanics are unit-tested without full JavaFX rendering).
- Cleaner scaling for future features (new UI widgets or new enemy/tower classes can evolve independently).

4. OOP Principles Applied

Inheritance

- Tower is an abstract base class concrete towers extend it
 - ArcherTower, CannonTower, CrossbowTower, etc.
- Enemy is an abstract base class enemy variants extend it
 - SlimeEnemy, BatEnemy, DemonEnemy, etc.

Code example

```
public abstract class Tower implements Upgradable { ... }
public final class ArcherTower extends Tower { ... }

public abstract class Enemy implements Damageable { ... }
public final class DemonEnemy extends Enemy { ... }
```

- Why it is important?
 - Shared fields and logic (position, stats, cooldowns, movement, health, etc.) are implemented once in base classes.
 - Reduces duplication and keeps balancing changes centralized.
 - New tower/enemy variants can be added quickly with minimal code.

Interface

- Upgradable interface implemented by Tower
- Damageable interface implemented by Enemy

Code example

```
public interface Upgradable {  
    void upgrade();  
}  
  
public interface Damageable {  
    void takeDamage(int amount);  
}
```

- Why it is important?
 - Defines behavior contracts independent of concrete class.
 - Encourages consistent APIs for upgrade and damage interactions.
 - Supports future extensibility (other entities can become upgradeable/damageable without changing consumers).

Polymorphism

- Runtime dispatch through base-class references
 - GameManager stores List<Tower> and List<Enemy>
 - Calls tower.update(...) and enemy.update(...) without needing subtype-specific branching
- Tower subclasses override update(...) to use their own projectile sprite/behavior.

Code example

```
for (Tower tower : activeTowers) {  
    tower.update(activeEnemies, activeProjectiles);  
}
```

```
@Override  
public void update(List<Enemy> enemies, List<Projectile> activeProjectiles) {  
    updateProjectileAttack(enemies, activeProjectiles, CANNON_PROJECTILE_SPRITE);  
}
```

- Why it is important?
 - Same method call, different runtime behavior by tower type.
 - Keeps the combat loop clean and open for extension.
 - Avoids large if/else or switch blocks for every subtype behavior.

Access Modifiers

- **private (encapsulation, internal invariants)**
 - e.g., GameManager state fields (playerMoney, baseHealth, timeRemaining, etc.)
 - helper methods hidden from external misuse (isPlacementRequestValid, isTileOccupiedByTower)
- **protected (shared subclass access)**
 - e.g., Tower and Enemy combat/state fields used by subclasses for specialization
- **public (external API)**
 - game-facing operations like update, getters/setters, placeTower, tryUpgradeTower

Code example

```
// GameManager
private int playerMoney;
public boolean placeTower(int row, int col) { ... }

// Tower
protected int damage;
```

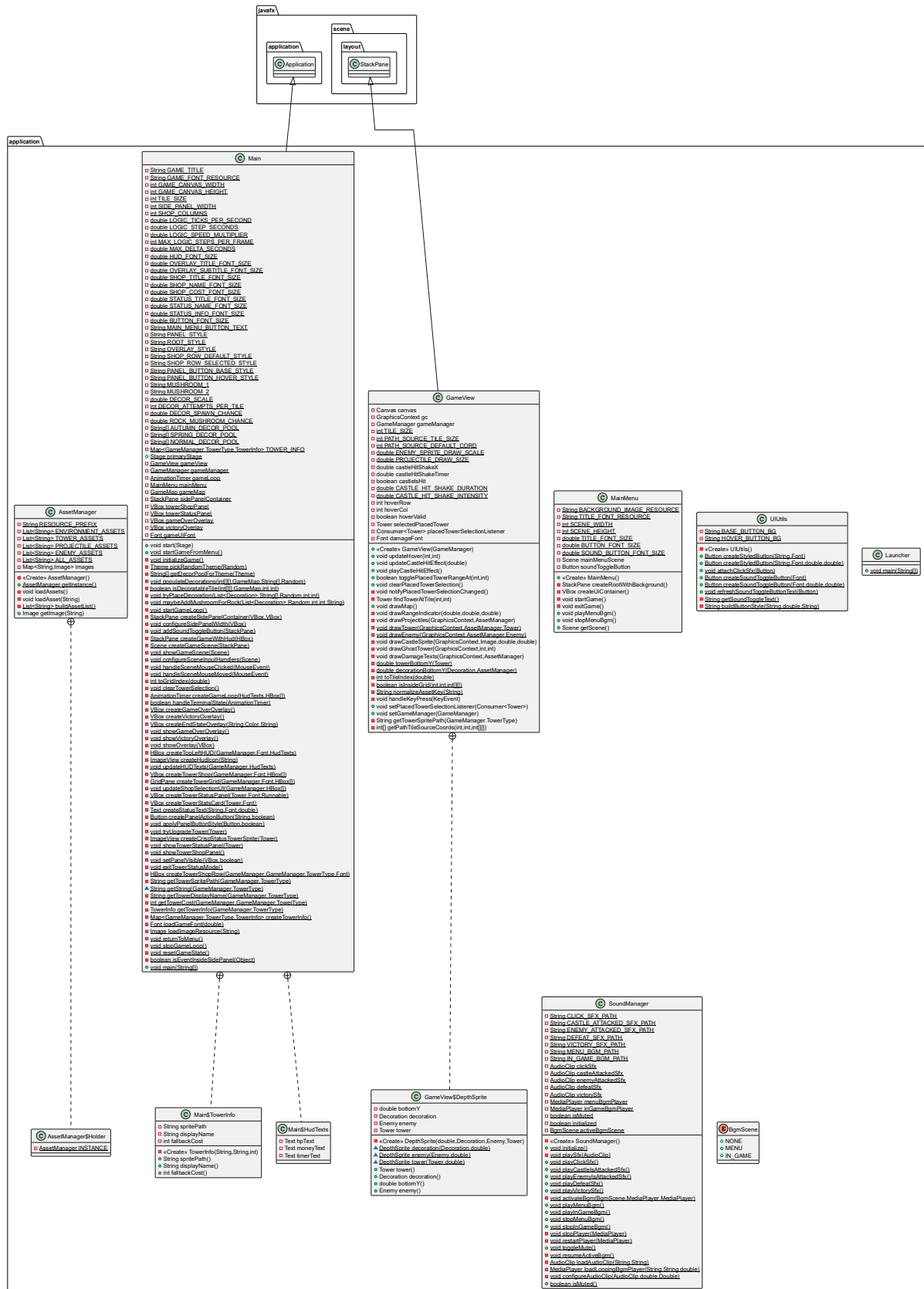
- Why it is important?
 - Protects critical game state from unsafe direct manipulation.
 - Gives controlled extension points to child classes.
 - Exposes only what other modules (UI/tests) must use.

5. JUnit Testing Strategy

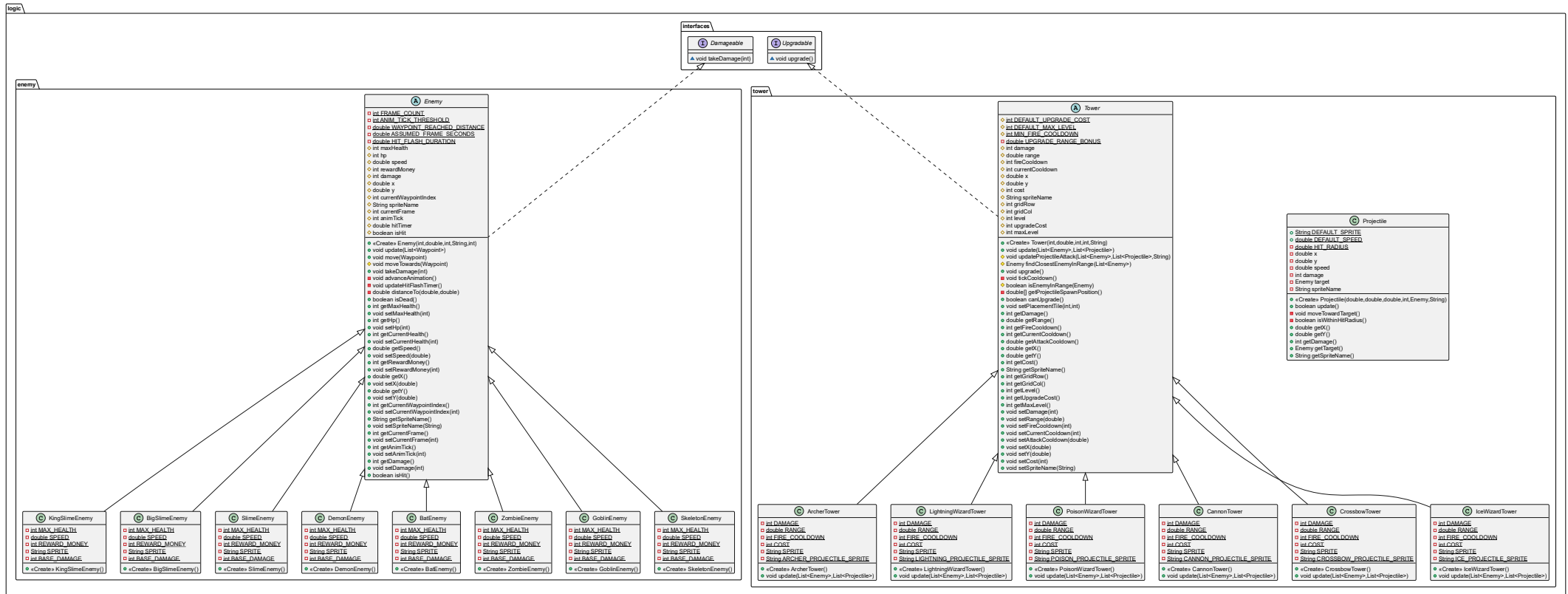
The src/test/java/logic suite focuses on **core gameplay logic** (high-value, deterministic units), not heavy JavaFX rendering

- **TowerTest:** targeting priority, cooldown behavior, upgrade rules, projectile sprite correctness.
- **EnemyTest:** movement/path following, hit flash timing, animation stepping, variant stats.
- **ProjectileTest:** movement, hit radius, overshoot protection, default sprite fallback.
- **GameMapTest:** buildability constraints, waypoint coordinate generation, BFS path behavior.
- **PathGeneratorTest:** generated map validity, single border entry, castle reachability.
- **LevelLoaderTest:** map parsing and fallback behavior on malformed/missing input.
- **DamageTextTest:** fade/lifetime and opacity clamping.
- **GameManagerTest:** economy, placement validity, upgrades, spawn placement, reset flow, victory/game-over transitions.

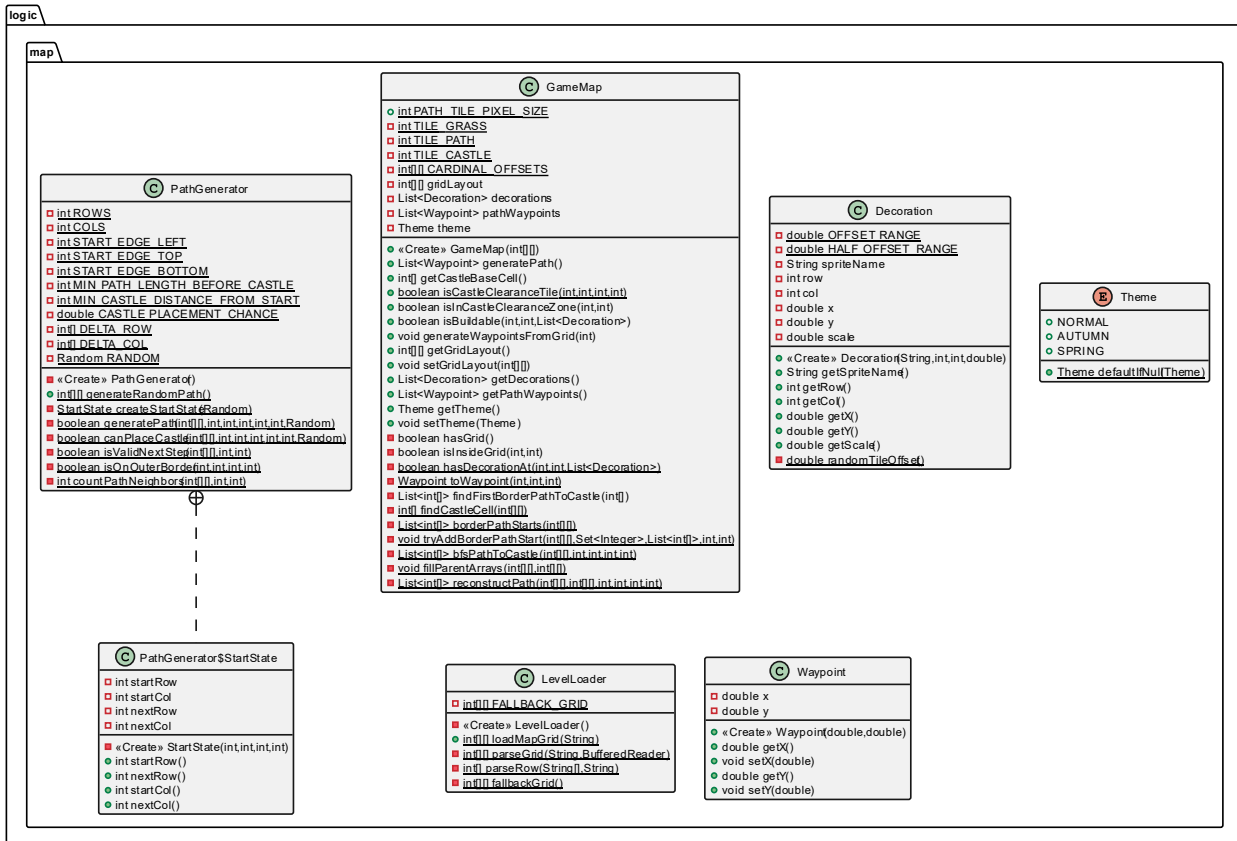
```
src/main/java/application
```



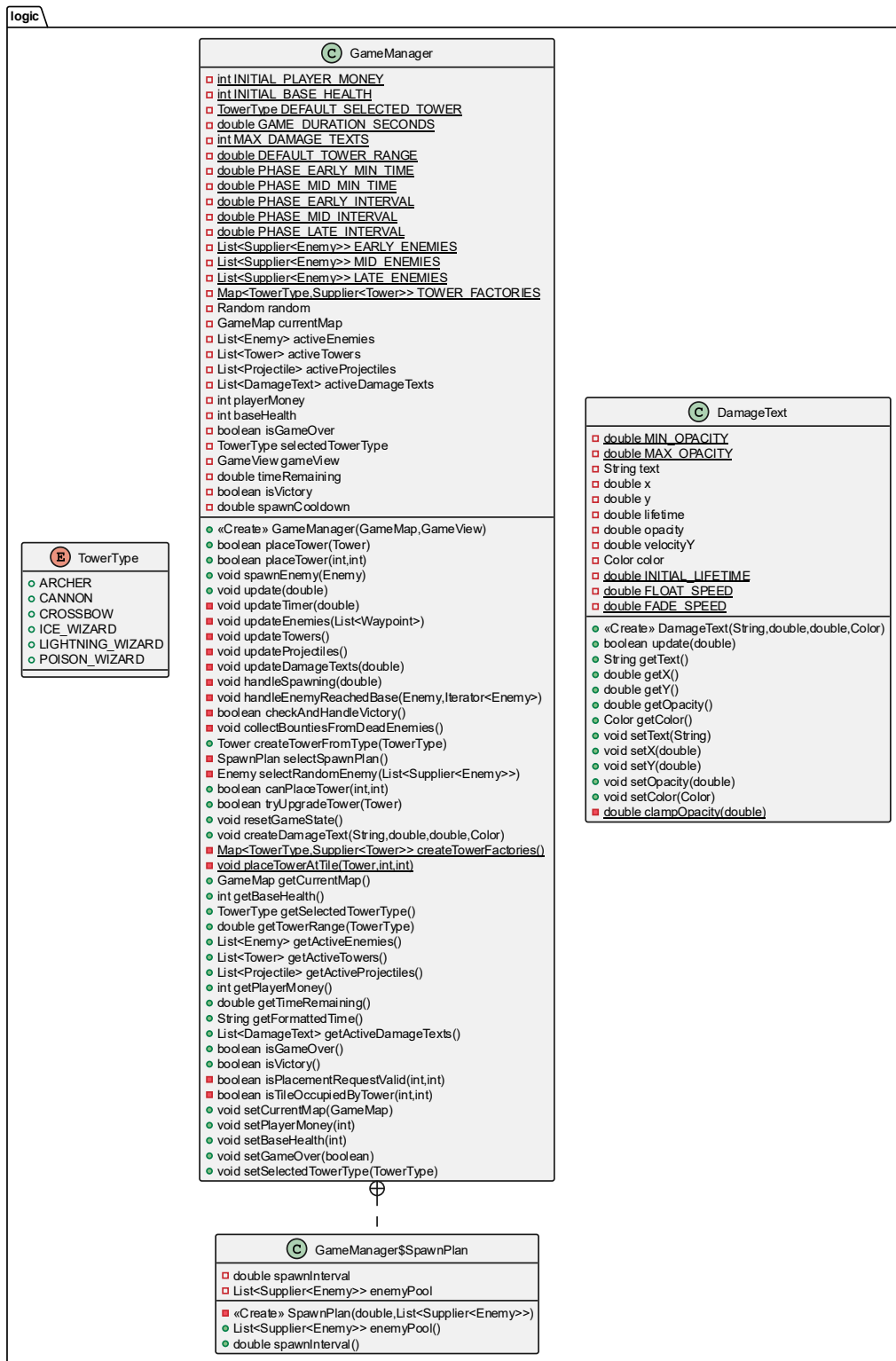
src/main/java/logic/enemy, src/main/java/logic/tower, and src/main/java/logic/interfaces



src/main/java/logic/map



src/main/java/logic/GameManager.java and src/main/java/logic/DamageText.java



7. Projects Links

- JavaDoc: [Click here to view JavaDoc](#)
- UML Diagram: [Click here to view full UML Diagram](#)